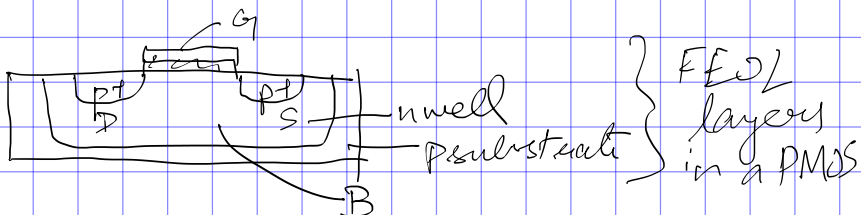


Physical Design/Layout

- * Foundries provide a document of devices that they can manufacture in a particular process technology. E.g. of semiconductor foundries are TSMC, Intel, Samsung, IHP, GF...
 - * Foundries lay some ground rules that "you" as a designer/layout engineer need to honour so that your design can be (reliably) fabricated.
 - * Layout - the means by which you "communicate" with the foundry. It is a composition of polygons/shapes in various "masks".
 - * A mask in the layout could correspond to a physical mask used by the foundry to "print" the shapes on the associated layer.
 - * Layers
 - FEOL - Front end of line
 - MEOL - Middle end of line
 - BEOL - Back end of line
- FEOL - layers used to construct the active devices such as n/pmos.



MEOL - layers used to "escape out" of active devices & to make short connections. e.g. within standard cells. Usually these layers are highly resistive due to the material used and the dimensions.

In Skywater 130, (li) & (licon) form the MEOL layers

BEOL - Interconnect layers used for medium and long connections. e.g. between gates and between modules.

metal 1 - metal 5 and the intermediate via layers form BEOL

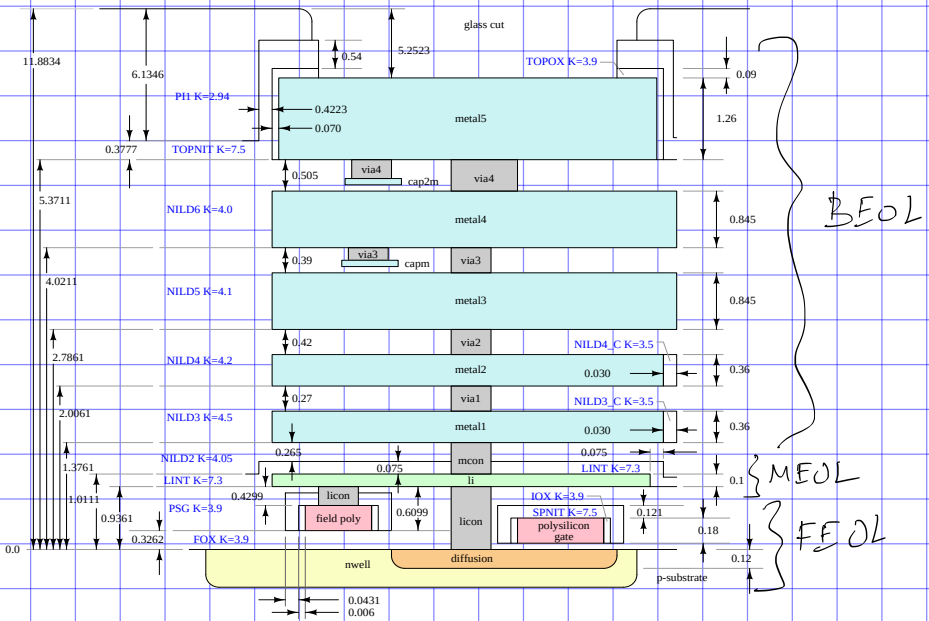
Commonly used layout format in the past: GDS2 - Graphic Design System.

Currently - OASIS - is the de facto format used in the industry.
 $\text{size of OASIS} \ll \text{size of GDS}$

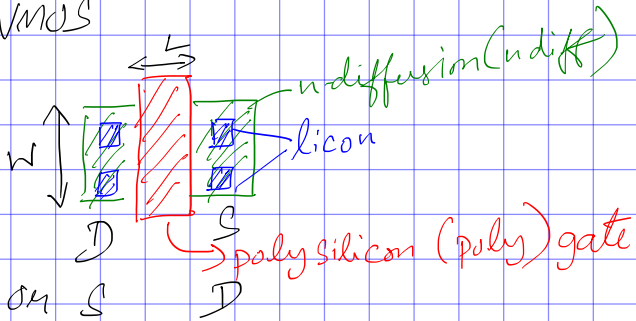
Tape out: Process of shipping GDS/OASIS files to the foundry.

In the past "tapes" of layout data were shipped off; hence the name.

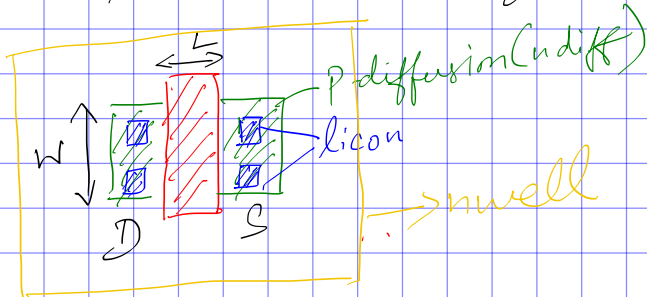
Skywater 130nm Process stack



Drawing NMOS



Source and Drain assignments in layout are interchangeable. Whenever Poly crosses a diffusion, a transistor is formed.



DRC - Design Rule Checks

Foundry defines certain geometric constraints that define physical limits of mask dimensions and other constraints that help reliably manufacture intended devices and their interconnections.

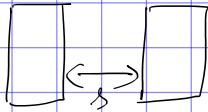
few rules

1.) min width/length rules.

$$L \geq L_{\min} = 0.15 \mu\text{m}$$

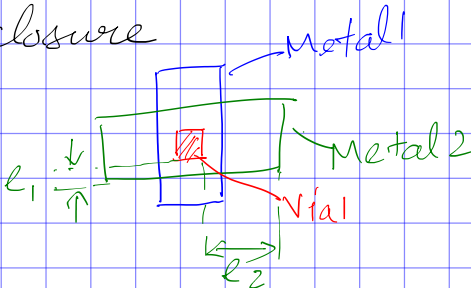
$$W \geq W_{\min} = 0.42 \mu\text{m}$$

2.) Spacing rule



$s \geq s_{\min}$ (layer specific and sometimes inter layer).

3.) Enclosure



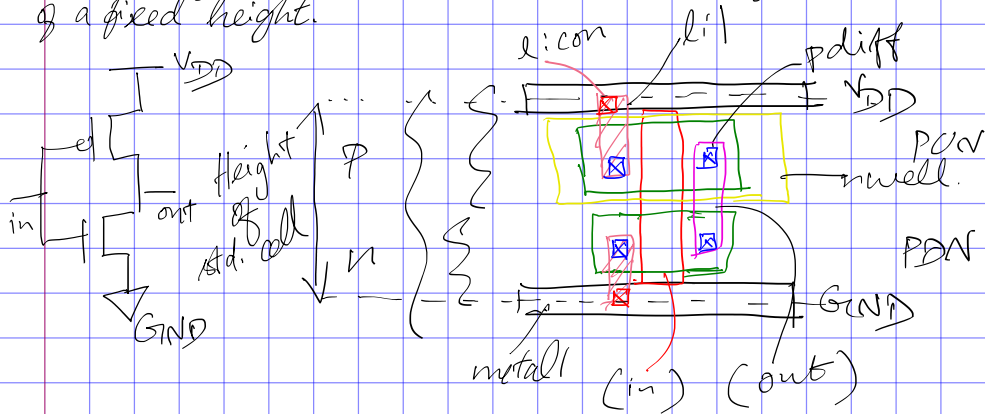
$$e_1 \geq e_{1\min}$$

$$e_2 \geq e_{2\min}$$

Standard cells

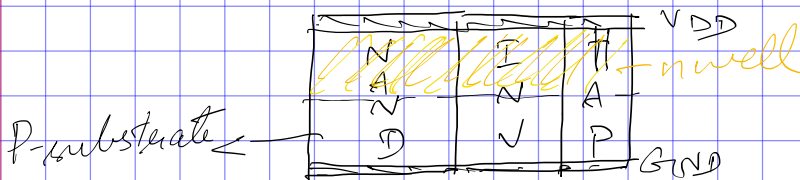
Standard cell is an interconnection of transistors that implements a logic function or a memory element. e.g. NOT, NAND, XOR, D-Flipflop...

They typically have an associated schematic/netlist and a layout. The cell layouts are typically fixed height or in modern processes integer multiple of a fixed height.



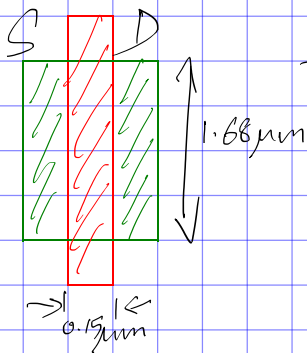
Well taps or ties are special cells used to connect the wells/substrate to corresponding power/ground.

⇒ multiple adjacent devices share the same tap.

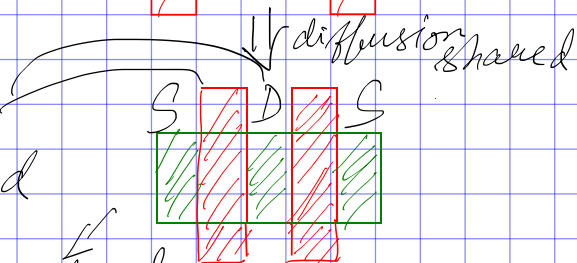
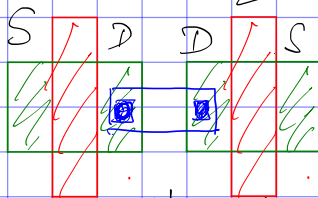


Diffusion Sharing

Consider a nmos with $\frac{W}{L} = \frac{1.68\mu m}{0.15\mu m}$



- This device can be folded using two NMOS of $\frac{W}{L} = \frac{0.84\mu m}{0.15\mu m}$

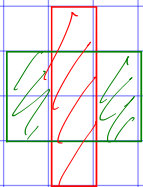


each of these poly are called fingers.

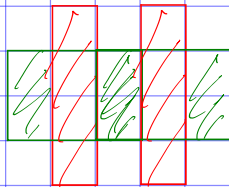
"nf" is the parameter used to specify number of fingers.

reduced C_{self} since junction cap is smaller.

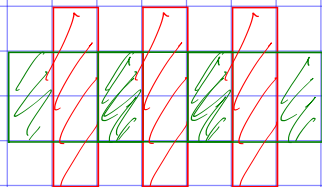
nf = 1



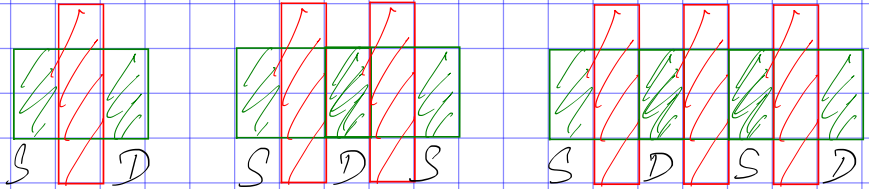
nf = 2



nf = 3



To match the parasitic between schematic and layout, n_f is a parameter used in Schematic simulation. For schematic as a convention the first diffusion on the left is considered source

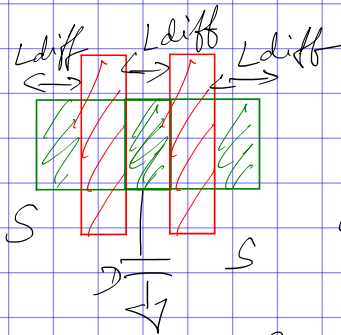


n_s - number of source regions
 n_d - number of drain regions

n_f	n_s	n_d
1	1	1
2	2	1
3	2	2
4	3	2
$\vdots$		

$$n_s = \left\lfloor \frac{n_f + 2}{2} \right\rfloor$$

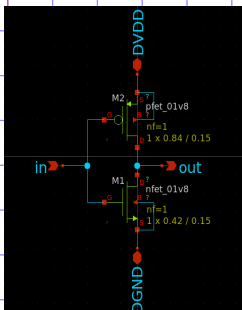
$$n_d = \left\lfloor \frac{n_f + 1}{2} \right\rfloor$$



$$C_{\text{drain bottom}} = C_j \times \frac{W_p}{n_f} \times L_{\text{diff}} \times n_d$$

$$C_{\text{source bottom}} = C_j \times \frac{W_p}{n_f} \times L_{\text{diff}} \times n_s$$

$$C_{\text{sidewall drain}} = n_d \times (2L_{\text{diff}} + 2\frac{W_p}{n_f})$$



$\xrightarrow{\text{PMOS}}$
 XM2 out in DVDD DVDD sky130 fd pr_pfet_01v8 L=0.15 W=0.84 nf=1
 + ad= int((nf+1)/2) * W/nf * 0.29
 + as= int((nf+2)/2) * W/nf * 0.29
 + pd= 2*int((nf+1)/2) * (W/nf + 0.29)
 + ps= 2*int((nf+2)/2) * (W/nf + 0.29)
 + nrd= 0.29 / W' nrs= 0.29 / W' sa=0 sb=0 sd=0 mult=1 m=1
 XM1 out in DGND DGND sky130 fd pr_nfet_01v8 L=0.15 W=0.42 nf=1
 + ad= int((nf+1)/2) * W/nf * 0.29
 + as= int((nf+2)/2) * W/nf * 0.29
 + pd= 2*int((nf+1)/2) * (W/nf + 0.29)
 + ps= 2*int((nf+2)/2) * (W/nf + 0.29)
 + nrd= 0.29 / W' nrs= 0.29 / W' sa=0 sb=0 sd=0 mult=1 m=1
 $\xrightarrow{\text{NMOS}}$

$$\frac{W_m}{L_m} = \frac{0.42 \mu}{0.15 \mu}$$

$$1.) C_{gu} = C_{g \text{ unit}} C_{qf}$$

$$= \frac{C_{ox} \times F_{min} \times W_{min}}{0.5 \text{ fF}}$$

$$C_g = C_{gu} \times \left(\frac{W}{W_{min}} \right)$$

$$2.) C_{jn} = \frac{A_{diff} \times W_{min} \times C_j}{0.167 \text{ fF}}$$

$$C_{jact} = C_{jn} \times \left(\frac{W}{W_{min}} \right)$$

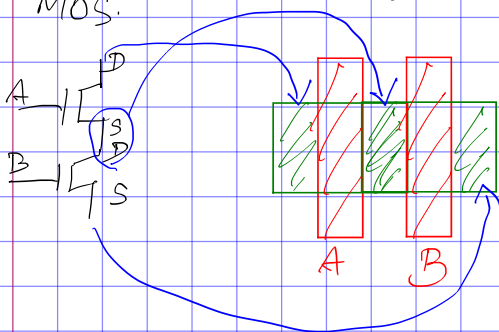
$$3.) C_{jsw} = C_{jsw} (2 \times 0.42 \mu + 2 \times 0.29 \mu)$$

$$= (0.2 + 0.142) \text{ fF}$$

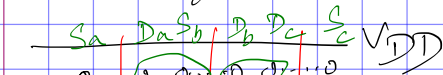
$$C_{jswact} = 0.2 + 0.142 \times \frac{W}{W_{min}}$$

$$4.) \text{Converlap} = \frac{W}{W_{min}} \times 0.2 \text{ fF}$$

Two series or parallel connect MOS can share diffusion just like a multifingered MOS.



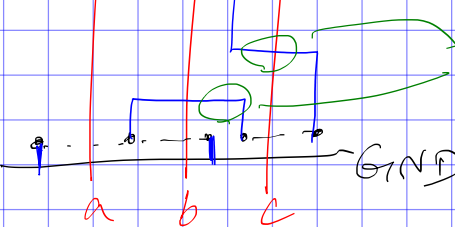
Stick diagram



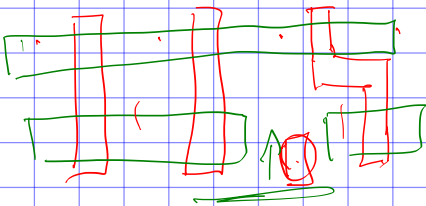
a b c

↓

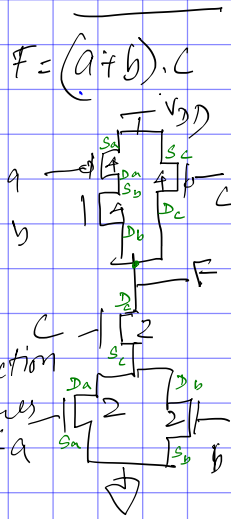
no connection
hence
requires
breaks



length of wire
longer because
of diffusion
break.



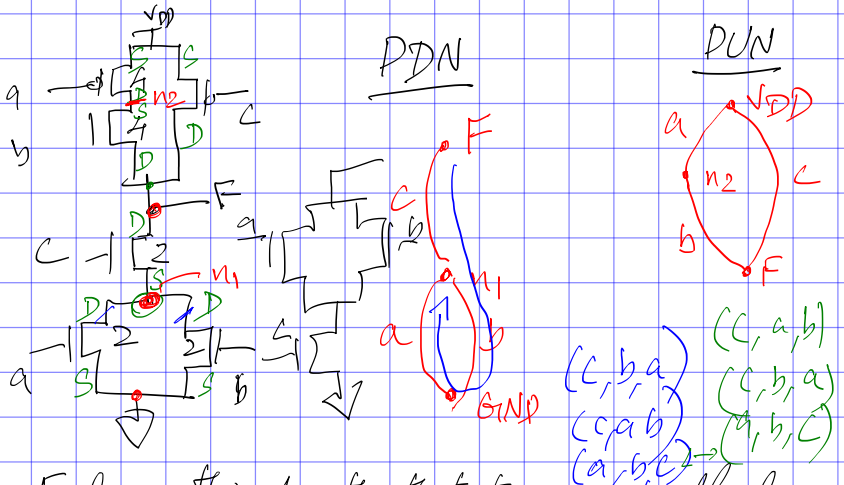
more diffusion
sharing
lesser are the
wire parasitic.



$$F = (a + b) \cdot c$$

$$F = (a+b) \cdot C$$

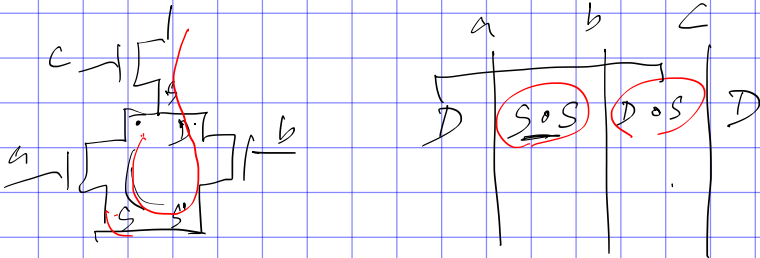
Graph: all acts in the schematic are vertices and transistors edges.



Euler path: A path that transverses all edges exactly once.

Try and find the same Euler path in both PUN & PDN.

e.g. (a, b, c) in the above graph.



$$F = (A \cdot (D + E)) + B \cdot C$$

Find Euler path for F.
that minimizes diffusion breaks.

